

CARBON MARKETS CCUS AND DIRECT AIR CAPTURE

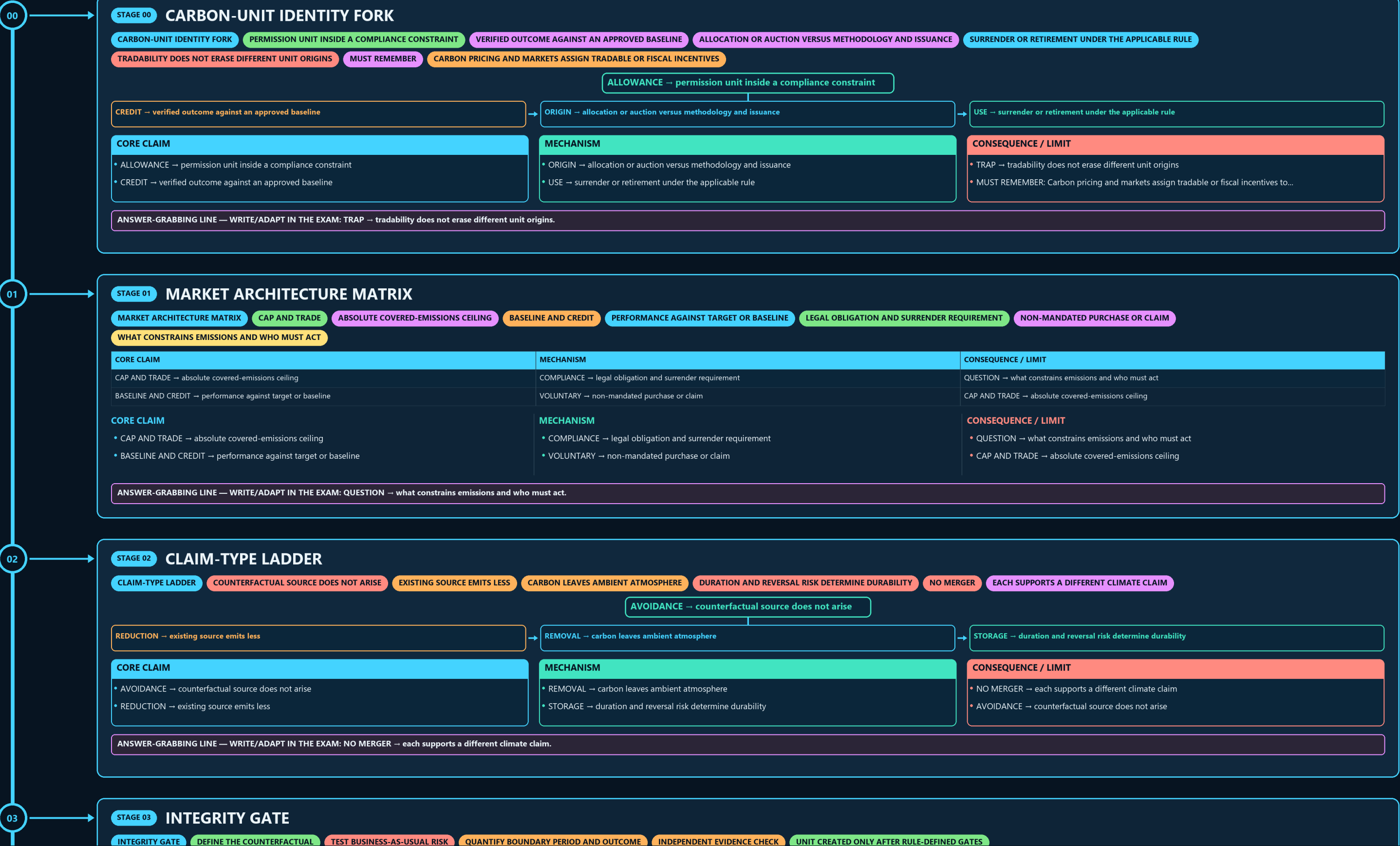
CARBON-UNIT IDENTITY FORK → MARKET ARCHITECTURE MATRIX → CLAIM-TYPE LADDER → INTEGRITY GATE → CREDIT LIFECYCLE RAIL → ACCOUNTING-RISK FIREWALL

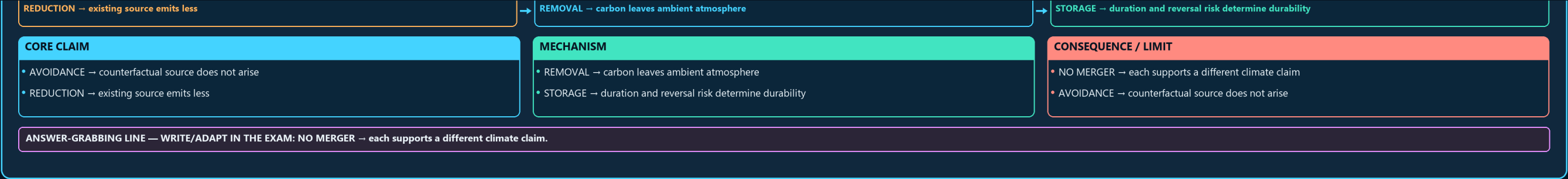
Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

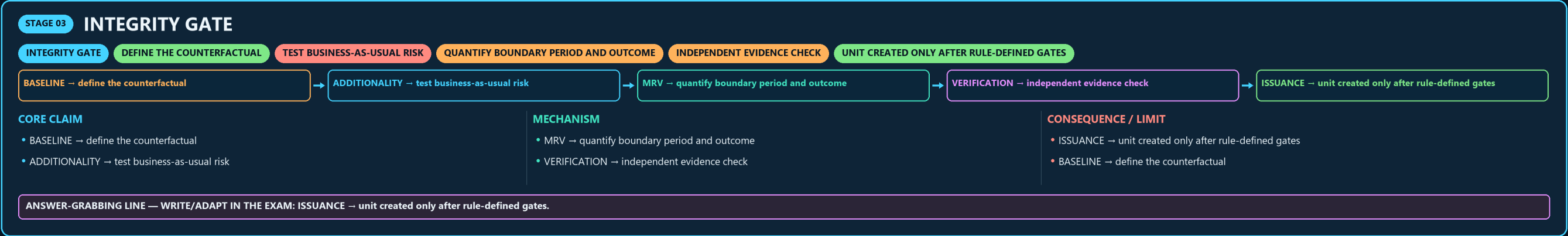
Approval: FALSE • Pending user review • source generation g5 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

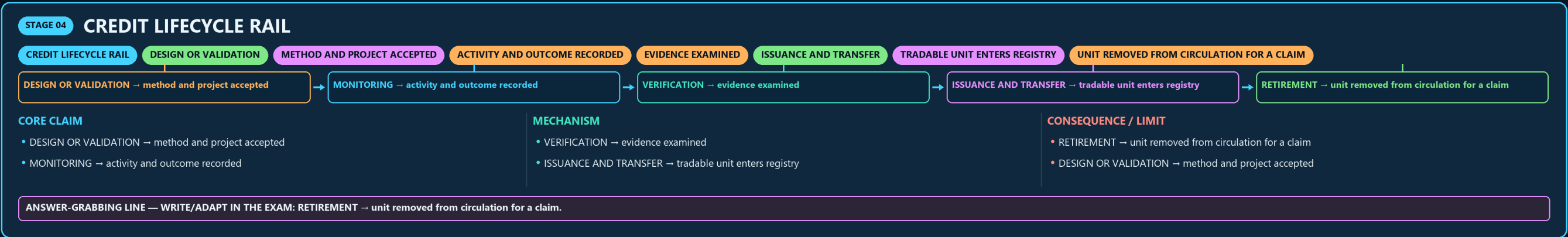




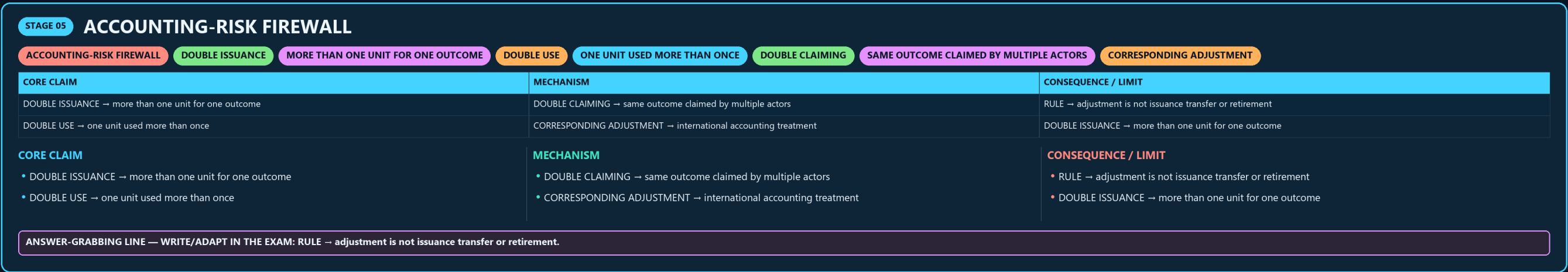
03



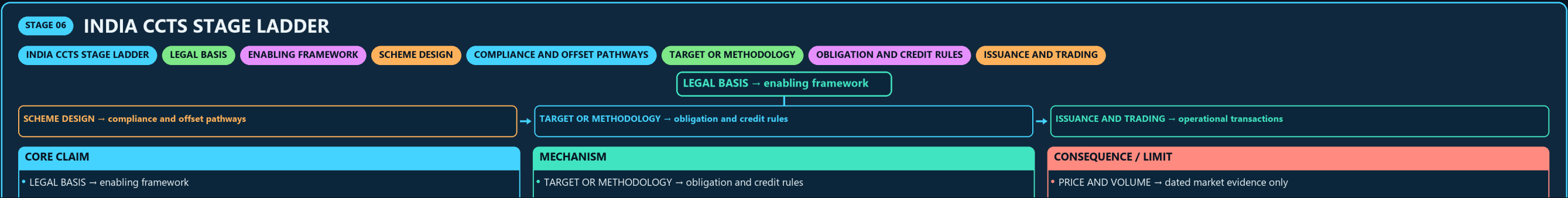
04



05



06



06

STAGE 06

INDIA CCTS STAGE LADDER

INDIA CCTS STAGE LADDERLEGAL BASISENABLING FRAMEWORKSCHEME DESIGNCOMPLIANCE AND OFFSET PATHWAYS

TARGET OR METHODOLOGYOBLIGATION AND CREDIT RULES

ISSUANCE AND TRADING

LEGAL BASIS → enabling framework

SCHEME DESIGN → compliance and offset pathways

TARGET OR METHODOLOGY → obligation and credit rules

ISSUANCE AND TRADING → operational transactions

CORE CLAIM

LEGAL BASIS → enabling framework

SCHEME DESIGN → compliance and offset pathways

MECHANISM

TARGET OR METHODOLOGY → obligation and credit rules

ISSUANCE AND TRADING → operational transactions

CONSEQUENCE / LIMIT

PRICE AND VOLUME → dated market evidence only

LEGAL BASIS → enabling framework

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PRICE AND VOLUME → dated market evidence only.

07

STAGE 07

CCUS CHAIN

CCUS CHAINCONCENTRATED CARBON DIOXIDE STREAMCAPTURE AND CONDITIONINGSEPARATION AND PREPARATIONPIPELINE SHIP ROAD OR OTHER ROUTEUTILISATION OR STORAGEDESTINATION DETERMINES DURABILITYMONITORING AND CLOSURE

SOURCE → concentrated carbon dioxide stream

CAPTURE AND CONDITIONING → separation and preparation

TRANSPORT → pipeline ship road or other route

UTILISATION OR STORAGE → destination determines durability

MONITORING AND CLOSURE → permanence and responsibility

CORE CLAIM

SOURCE → concentrated carbon dioxide stream

CAPTURE AND CONDITIONING → separation and preparation

MECHANISM

TRANSPORT → pipeline ship road or other route

UTILISATION OR STORAGE → destination determines durability

CONSEQUENCE / LIMIT

MONITORING AND CLOSURE → permanence and responsibility

SOURCE → concentrated carbon dioxide stream

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRANSPORT → pipeline ship road or other route.

08

STAGE 08

UTILISATION-STORAGE TEST

UTILISATION-STORAGE TESTPRODUCT USEHOW LONG IS CARBON RETAINEDPROCESS USEDOES LATER RELEASE OCCURGEOLOGICAL STORAGECONTAINMENT AND MONITORINGREVERSAL RISK AND REMEDIATION

PRODUCT USE → how long is carbon retained

PROCESS USE → does later release occur

GEOLOGICAL STORAGE → containment and monitoring

LEAKAGE → reversal risk and remediation

CORE CLAIM

PRODUCT USE → how long is carbon retained

PROCESS USE → does later release occur

MECHANISM

GEOLOGICAL STORAGE → containment and monitoring

LEAKAGE → reversal risk and remediation

CONSEQUENCE / LIMIT

NET BENEFIT → full lifecycle accounting required

PRODUCT USE → how long is carbon retained

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PRODUCT USE → how long is carbon retained.

09

STAGE 09

CAPTURE TECHNOLOGY COMPARISON

CAPTURE TECHNOLOGY COMPARISONPOINT SOURCE CCUSCARBON INTERCEPTED BEFORE RELEASECARBON SEPARATED FROM AMBIENT AIRBIOGENIC CARBON PLUS CAPTURE AND STORAGEREMOVAL TESTATMOSPHERIC ORIGIN PLUS DURABLE STORAGELIFECYCLE TEST

CORE CLAIM

POINT SOURCE CCUS → carbon intercepted before release

DAC → carbon separated from ambient air

MECHANISM

BECCS → biogenic carbon plus capture and storage

REMOVAL TEST → atmospheric origin plus durable storage

CONSEQUENCE / LIMIT

LIFECYCLE TEST → energy land transport and reversal

CLOSE DISTINCTION: Allowance is not offset, avoidance is not removal, capture is not...

CORE CLAIM

POINT SOURCE CCUS → carbon intercepted before release

DAC → carbon separated from ambient air

MECHANISM

BECCS → biogenic carbon plus capture and storage

REMOVAL TEST → atmospheric origin plus durable storage

CONSEQUENCE / LIMIT

LIFECYCLE TEST → energy land transport and reversal

CLOSE DISTINCTION: Allowance is not offset, avoidance is not removal, capture is not...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Allowance is not offset, avoidance is not removal, capture is not.

09

STAGE 09

CAPTURE TECHNOLOGY COMPARISON

CAPTURE TECHNOLOGY COMPARISON

POINT SOURCE CCUS

CARBON INTERCEPTED BEFORE RELEASE

CARBON SEPARATED FROM AMBIENT AIR

BIOGENIC CARBON PLUS CAPTURE AND STORAGE

REMOVAL TEST

ATMOSPHERIC ORIGIN PLUS DURABLE STORAGE

LIFECYCLE TEST

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
POINT SOURCE CCUS → carbon intercepted before release	BECCS → biogenic carbon plus capture and storage	LIFECYCLE TEST → energy land transport and reversal
DAC → carbon separated from ambient air	REMOVAL TEST → atmospheric origin plus durable storage	CLOSE DISTINCTION: Allowance is not offset, avoidance is not removal, capture is not...

CORE CLAIM

POINT SOURCE CCUS → carbon intercepted before release

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MECHANISM

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REMOVAL TEST → atmospheric origin plus durable storage

CONSEQUENCE / LIMIT

LIFECYCLE TEST → energy land transport and reversal

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ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Allowance is not offset, avoidance is not removal, capture is not.

10

STAGE 10

NET-ZERO MITIGATION HIERARCHY

NET-ZERO MITIGATION HIERARCHY

PREVENT UNNECESSARY EMISSIONS

DECARBONISE EXISTING ACTIVITIES

ADDRESS HARD-TO-ABATE POINT SOURCES

COUNTERBALANCE GENUINELY RESIDUAL EMISSIONS

NET BALANCE NEEDS ONE BOUNDARY AND PERIOD

EVIDENCE LIMIT

FIX SYSTEM BOUNDARY, BASELINE, UNIT

REDUCE → decarbonise existing activities

CAPTURE → address hard-to-abate point sources

REMOVE → counterbalance genuinely residual emissions

AVOID → prevent unnecessary emissions

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
AVOID → prevent unnecessary emissions REDUCE → decarbonise existing activities	CAPTURE → address hard-to-abate point sources REMOVE → counterbalance genuinely residual emissions	VERIFY → net balance needs one boundary and period EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Fix system boundary, baseline, unit,...

MECHANISM / VERDICT — EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Fix system boundary, baseline, unit,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Fix system boundary, baseline, unit,.

11

STAGE 11

CARBON INSTRUMENTS ANSWER SPINE

CARBON INSTRUMENTS ANSWER SPINE

UNIT MARKET AND CLAIM TYPE

BASELINE MRV ISSUANCE TRANSFER RETIREMENT

CAPTURE TRANSPORT UTILISATION OR STORAGE

PERMANENCE LEAKAGE LIFECYCLE AND DOUBLE COUNTING

CURRENT RULE PRICE VOLUME COST AND CAPACITY SOURCE

CLASSIFY → unit market and claim type

TRACE → baseline MRV issuance transfer retirement

FOLLOW → capture transport utilisation or storage

TEST → permanence leakage lifecycle and double counting

AUDIT → current rule price volume cost and capacity source

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
CLASSIFY → unit market and claim type TRACE → baseline MRV issuance transfer retirement	FOLLOW → capture transport utilisation or storage TEST → permanence leakage lifecycle and double counting	AUDIT → current rule price volume cost and capacity source CLASSIFY → unit market and claim type

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: AUDIT → current rule price volume cost and capacity source.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT

DAC'S FUNDAMENTAL TECHNICAL CHALLENGE IS THERMODYNAMIC

ATMOSPHERIC CO₂ CONCENTRATION

POLICY IMPLICATION

BECAUSE DAC DELIVERS GENUINE ATMOSPHERIC REMOVAL (UNLIKE

GLOBAL CCUS AND DAC DEPLOYMENT CAPACITY REMAINS SMALL RELATIVE

THE INDIAN CARBON MARKET'S SPECIFIC SECTORAL COVERAGE, TARGET-SETTING METHODOLOGY

MRV-SYSTEM ROBUSTNESS FOR BOTH INDIA'S DOMESTIC CARBON MARKET AND

OPTIONAL SOURCE DEPTH	ADVANCED DEBATE	LEGACY / NUANCE
CAUTION: DAC's fundamental technical challenge is thermodynamic: atmospheric CO₂ concentration CAUTION: Policy implication: because DAC delivers genuine atmospheric removal (unlike CAUTION: Global CCUS and DAC deployment capacity remains small relative to the scale IPCC CAUTION: The Indian Carbon Market's specific sectoral coverage, target-setting methodology and	CAUTION: MRV-system robustness for both India's domestic carbon market and international India's carbon market has a two-track design: a compliance mechanism (building on PAT, Standard fossil-source CCUS is an emissions-avoidance technology; genuine atmospheric DAC's higher cost/energy-intensity relative to CCUS stems from the much lower CO₂	Net-zero frameworks require genuine removal technologies (DAC/BECCS/nature-based sinks) NOT: All CCUS applications, including standard fossil-power-plant CCUS, deliver negative NOT: India's carbon market is a single-track, purely compliance-based system. - It has a NOT: DAC is currently deployed at a scale comparable to global CCUS capacity. - DAC remains

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.